

Lecture 2: Thermodynamic Equilibrium: Special Limits and Algebraic Structures

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You can find these lecture notes and some exercises at:
<https://olallacastroalvaredo.weebly.com/pentegost-school-25.html>



AS I TOLD YOU YESTERDAY...

Thermodynamic Bethe Ansatz Equations

- The minimisation constraint combined with the relationship between ρ_s and ρ_p gives the **TBA equations**

Thermodynamic Bethe Ansatz Equations

$$\varepsilon_a(\theta) = m_a \beta \cosh \theta - \sum_{b=1}^N \varphi_{ab} * L_b(\theta), \quad a = 1, \dots, N, \quad \beta := \frac{1}{T}$$

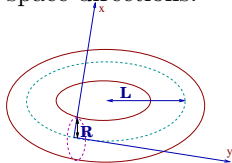
$L_a(\theta) = \pm \log(1 \pm e^{-\varepsilon_a(\theta)})$ for fermions (+) and bosons (-).

- They are **coupled integral equations** for $\varepsilon_a(\theta)$.
- Even for complicated theories, they can be easily solved numerically through a recursive procedure.
- For free theories $\varphi_{ab}(\theta) = 0$ and $\varepsilon_a(\theta) = m_a \beta \cosh \theta$ are simply the free “on-shell” energies.
- The **extremal free energy** is simply

$$f(\beta) = -\frac{1}{\beta} \sum_{a=1}^N m_a \int_{-\infty}^{\infty} \frac{d\theta}{2\pi} \cosh \theta L_a(\theta)$$

Relation to IQFT on a Torus

- Consider our QFT on a torus of circumferences L and R . We may now quantise the system in two possible ways depending on the choice of the time and space directions.



- Taking $L \rightarrow \infty$ as time direction and keeping R finite we have a compactified system whose partition function is led by the term $e^{-E_0(R)L}$ where $E_0(R)$ is the **ground state energy**.
- Exchanging the roles of time and space we have instead a theory where time is periodic and space infinite. If we identify $R = T^{-1} = \beta$ then we can think of this as a QFT at **finite temperature**.
- The partition function is now dominated by the term $e^{-RLf(R)}$ where $f(R)$ is the **free energy per unit length**.

Ground State Energy and Central Charge

- It follows from the previous slide that

$$E_0(R) = Rf(R) = - \sum_{a=1}^N m_a \int_{-\infty}^{\infty} \frac{d\theta}{2\pi} \cosh \theta L_a(\theta)$$

- The limit $R \rightarrow 0$ is the **ultraviolet or high energy limit**. In this limit we expect to approach the **underlying CFT** and so the ground state energy is related to the **central charge**:

$$\lim_{R \rightarrow 0} E_0(R) = -\frac{\pi c_{\text{eff}}}{6R} \quad \text{with} \quad c_{\text{eff}} = c - 24\Delta$$

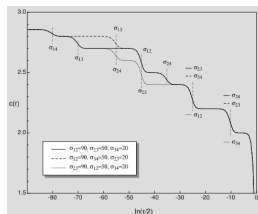
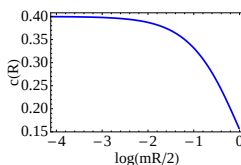
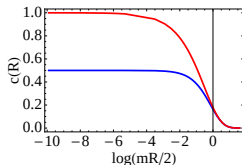
- This provides a strong consistency check of the S -matrix.
- For generic values of R we may define a **scaling function**:

TBA Scaling Function

$$c(R) = \frac{3R}{\pi^2} \sum_{a=1}^N m_a \int_{-\infty}^{\infty} d\theta L_a(\theta) \cosh \theta.$$

Features of the Scaling Function

- The scaling function $c(R)$ is one of the most studied objects within the TBA approach. It admits various interpretations: as an **“off-critical” Casimir energy** and as a measure of **the number of degrees of freedom** which are excited at a particular energy scale $1/R$.
- For massive QFTs $\lim_{R \rightarrow \infty} c(R) = 0$ and $\lim_{R \rightarrow 0} c(R) = c_{\text{eff}}$.
- The function $c(R)$ smoothly interpolates between these two values and, at least for unitary theories, is **monotonically decreasing** (from the UV to IR).

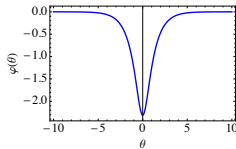
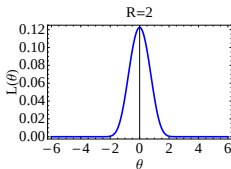
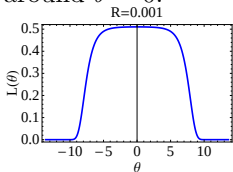


Constant TBA Equations

- The UV limit of the TBA equations can often be treated **analytically and exactly**.
- The equations resulting in this limit are known as **constant TBA equations**. They are coupled algebraic equations which encode beautiful **algebraic structures**.
- Recall once more the TBA equations:

$$\varepsilon_a(\theta) = m_a R \cosh \theta - \sum_{a=1}^{\infty} (\varphi_{ab} * L_b)(\theta), \quad a = 1, \dots, N$$

- As $R \rightarrow 0$ the functions $\varepsilon_a(\theta)$, $L_a(\theta)$ are **constant** for a wide range of values of θ . The kernels $\varphi_{ab}(\theta)$ are **strongly peaked** around $\theta = 0$.



- Then, the TBA equations become simply:

Constant TBA Equations

$$\varepsilon_a + \sum_{b=1}^N N_{ab} L_b = 0 \quad \text{with} \quad N_{ab} = \frac{1}{2\pi} \int_{-\infty}^{\infty} d\theta \varphi_{ab}(\theta)$$

- The constant TBA equations can be exactly solved for known N_{ab} . For many theories, N_{ab} encodes **algebraic information** about the structure of the S -matrices.
- For minimal Toda field theory: $N_{ab} = \delta_{ab} - 2K_{ab}^{-1}$ where K_{ab} is the **Cartan matrix** associated to a **simply-laced algebra**.
- Taking the infinite temperature limit of the TBA equations and the formula for the scaling function, it is possible to show:

$$c_{\text{eff}} = \frac{6}{\pi^2} \sum_{a=1}^N \mathcal{L} \left(\frac{1}{1+e^{\varepsilon_a}} \right) \quad \text{with} \quad \mathcal{L}(x) = \frac{1}{2} \int_0^x dy \left(\frac{\ln y}{y-1} - \frac{\log(1-y)}{y} \right)$$

where $\mathcal{L}(x)$ is **Roger's dilogarithm function**.

- A Y-system is an **alternative representation** of the TBA equations. The key functions are $Y_a(\theta) = e^{-\varepsilon_a(\theta)}$.
- This is best illustrated with a simple example: consider a theory with kernel $\varphi(\theta) = 2\text{sech}\theta$.
- The TBA-equation is simply $\xi(\theta) + (\varphi * L)(\theta) = 0$ for $\xi(\theta) = \varepsilon(\theta) - mR \cosh \theta$. Let us **Fourier-transform** this equation:

$$\tilde{\xi}(\omega) + \tilde{\varphi}(\omega)\tilde{L}(\omega) = 0 \quad \text{with} \quad \tilde{\varphi}(\omega) = \frac{1}{\pi} \int_{-\infty}^{\infty} d\theta \frac{e^{i\theta\omega}}{\cosh \theta} = \text{sech} \frac{\pi\omega}{2}$$

- Where the “tildes” indicate Fourier-transformed functions. So the equation becomes $\cosh \frac{\pi\omega}{2} \tilde{\xi}(\omega) + \tilde{L}(\omega) = 0$.
- Take now the inverse Fourier transform of this equation. It is possible to use the following property:

$$\int_{-\infty}^{\infty} d\omega e^{-i\theta\omega} \cosh \frac{\pi\omega}{2} \tilde{\xi}(\omega) = \frac{1}{2} \left(\xi(\theta - \frac{i\pi}{2}) + \xi(\theta + \frac{i\pi}{2}) \right).$$

Y-Systems: Example

- The TBA equations are now $\xi(\theta - \frac{i\pi}{2}) + \xi(\theta + \frac{i\pi}{2}) + 2L(\theta) = 0$
- The key property of these new equations is that the **driving term is cancelled out** in the combination $\xi(\theta - \frac{i\pi}{2}) + \xi(\theta + \frac{i\pi}{2})$!
- Exponentiating both sides and using the definition $Y(\theta) = e^{-\varepsilon(\theta)}$ we have $L(\theta) = \ln(1 + Y(\theta))$ and

Sinh-Gordon Y-System

$$Y(\theta - \frac{i\pi}{2})Y(\theta + \frac{i\pi}{2}) = (1 + Y(\theta))^2$$

- This is a **finite-difference equation** which contains the same information as the original integro-differential equation. The driving term is now “hidden” in the **asymptotic properties**.
- For many theories, it is possible to use the equations recursively to establish **periodicity** of the Y -functions. The period is related to the **conformal dimension of the perturbing field** as $Y(\theta + i\pi(1 - \Delta)^{-1}) = Y(\theta)$.

Beyond the Standard Construction

- In recent years, it has become clear that more general versions of these TBA equations can describe more general, **non-thermal states** of IQFT.
- The motivation comes from the study of **generalised Gibbs ensembles** and the understanding that for IQFTs they provide a description of **equilibration** as opposed to **thermalisation**.
- At the level of the TBA equations, the generalisation is simple: the driving term $m\beta \cosh \theta$ is replaced by a more general function $w(\theta)$. This is typically a sum over **one-particle eigenvalues of local and quasi-local conserved charges**.

$$w(\theta) = \sum_s (\beta_s m^s \cosh(s\theta) + \beta'_s m^s \sinh(s\theta)).$$

- In these GGEs we would like to be able to compute **average densities/currents of conserved quantities**, beyond the energy. The average density of an “even” conserved quantity q_s is

$$q_s = \int_{-\infty}^{\infty} d\theta \rho_p(\theta) h_s(\theta) \quad \text{with} \quad h_s(\theta) = m^s \cosh(s\theta)$$

Useful Relationships Going into GHD...

- As seen earlier, for fermionic theories we have

$$n(\theta) := \frac{\rho_p(\theta)}{\rho_s(\theta)} = \frac{1}{1 + e^{\varepsilon(\theta)}}, \quad L(\theta) = \ln(1 + e^{-\varepsilon(\theta)})$$

- The densities $\rho_p(\theta), \rho_s(\theta)$ are referred to as the **(quasi)particle and state (spectral) densities**.
- Their relationship $2\pi\rho_s(\theta) = p'(\theta) + 2\pi(\varphi * \rho_p)(\theta)$ is called the **constitutive relation**. From this equation, one can derive $2\pi\rho_p(\theta) = n(\theta)(p')^{\text{dr}}(\theta)$, where

$$f^{\text{dr}}(\theta) = f(\theta) + (\varphi * g)(\theta) \quad \text{with} \quad g(\theta) = n(\theta)f^{\text{dr}}(\theta)$$

defines the **dressing operation** which describes the change to one-particle eigenvalues due to the presence of interactions.

- We then have that

$$\mathbf{q}_s = \int_{-\infty}^{\infty} \frac{d\theta}{2\pi} n(\theta)(p')^{\text{dr}}(\theta)h_s(\theta) = \int_{-\infty}^{\infty} \frac{d\theta}{2\pi} n(\theta)p'(\theta)h_s^{\text{dr}}(\theta) = \int_{-\infty}^{\infty} \frac{dp(\theta)}{2\pi} n(\theta)h_s^{\text{dr}}(\theta)$$

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Constant TBA & Y-Systems

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